

Tab 1

I'll read through all three documents carefully before giving you a structured critique. Good — I have the full content of all three papers already in my context window from the document blocks. Let me now give you a thorough analysis.

Critical Assessment of the Generalized OF / EMPCA Note

Here is an honest gap analysis across five dimensions: mathematical completeness, connection to NIM/detector physics literature, signal processing generality, ML framing, and structure/presentation.

1. What the Note Does Well

Before the critique: the equivalence hierarchy in Section 6 is genuinely the most valuable part. Theorems 6.1 and 6.2, the whitened-space geometric interpretation, and the nested hierarchy (basic OF $\subset$ shifted OF $\subset$ low-rank EMPCA) are clean and correct. The autoencoder bridge in Section 7 is also solid and numerically verified.

2. Mathematical Gaps

Missing: Cramér-Rao Bound analysis. Yu's paper makes CRB optimality a central claim — the PCA/EMPCA method is argued to saturate the CRB for amplitude estimation under linear signal models. Your note states this verbally ("maximizing Gaussian likelihood") but never derives the Fisher information or shows the estimator variance achieves the bound. This is a critical omission for a NIM-style paper, where readers expect the words "minimum variance unbiased estimator" to be backed by the calculation. You should add:

$$I(A) = \frac{\partial^2}{\partial A^2} \log \mathcal{L} = \sum_k \frac{|S_k|^2}{J_k} = N_{\Phi}$$

and verify $\text{Var}(\hat{A}) = N_{\Phi}^{-1}$, matching the CRB.

Missing: convergence guarantees for EMPCA. Bailey's algorithm is iterative. Your note states the EM procedure converges but gives no rate, no conditions, and no reference to the known result that each iteration is non-increasing in χ^2 . Bailey (2012) proves monotone decrease; you should cite and state this, especially because Yu's hyperparameter section implicitly acknowledges convergence failure modes.

Incomplete: the time-shift section (6.7) is qualitative. The Taylor expansion argument in Proposition 6.1 is hand-wavy. You correctly identify that shifted pulses span $[s, s + \partial_t s]$,

$\partial_t^2 s, \dots)$, but you never bound the approximation error as a function of rank k and the RMS timing jitter σ_{t_0} . This would be the key quantitative result: how many EMPCA components do you need to capture a jitter of σ_{t_0} to within ϵ of the shifted-OF resolution? That calculation is missing and would make Section 6.7 publishable.

Missing: the non-identifiability / gauge freedom discussion is incomplete. You mention that rank- k EMPCA is defined only up to a unitary rotation within the subspace. This matters practically: how do you extract physically interpretable amplitudes from complex-valued EMPCA coefficients? The ridge regression stub in Section 3.5.2 points at this but is never developed. For a NIM paper, this is a critical workflow gap.

Missing: error propagation to energy resolution. The whole point for a microcalorimeter paper is energy resolution σ_E . You compute $\hat{A}$ and its variance, but you never propagate through the calibration chain (ADC $\rightarrow$ eV) to get a predicted σ_E in eV as a function of noise PSD components. Study B in Section 9 gestures at this but stops at a "relative sensitivity proxy" rather than an absolute number. Yu quotes $\sigma = 0.037$ in dimensionless units; CDMS/LAMCAL papers (Kurinsky, Fixsen et al.) quote FWHM in eV. Your note needs to close this loop.

3. Missing Connection to the NIM / Detector Physics Literature

This is the largest gap relative to what a NIM audience would expect.

Fowler et al. (2016, 2018) — "When Optimal Filtering Isn't" is cited by Yu but not engaged. Fowler's papers introduce optimal nonlinear analysis (ONA) and the nonlinear optimal filter (NLF) for microcalorimeters where the pulse shape is amplitude-dependent (saturation, quasiparticle recombination nonlinearity). Your note's Section 5 ("Non-linear feature extraction ResNet") is a stub. You should at minimum explain where the linear framework breaks down and why ResNet or NLF is the appropriate extension — this is the natural motivation for Section 5.

The Fixsen/Moseley group's pre-processing literature (baseline correction, glitch rejection, pile-up detection) is completely absent. In practice, EMPCA on raw microcalorimeter data fails unless you have clean baselines. The assumption of stationarity and zero-mean noise in Section 9.4 is stated but not defended against the real pathologies (slow drift, thermal fluctuations, cosmic-ray glitches).

CDMS / SuperCDMS literature (specifically the phonon optimal filter formalism in Kurinsky's thesis, which you do reference) is used only for the OF derivation. But Kurinsky also derives the multi-channel correlated OF, which your Section 2.3 leaves as a stub. You should fill this in — it is the direct physics motivation for Section 6.8.

Microcalorimeter-specific noise sources are treated in Section 9, but Table 4 and Table 6 are empty. The noise budget study (Study A) produces a beautiful figure (Fig. 3–4) but draws no conclusions because the tables are blank. A NIM reader expects numbers: at what frequency does SQUID noise dominate over Johnson noise? What is the crossover for your LAMCAL geometry?

4. Signal Processing Gaps

No discussion of the discrete-time / finite-record-length effects. The OF derivation assumes infinite records with exact DFT bins. In practice you have N -point DFTs, spectral leakage, and windowing. How does the PSD estimate (needed to set J_k) affect the OF bias and variance? This is discussed in the NMR and gravitational-wave literature but absent here.

No treatment of pile-up / overlapping pulses. This is a real operational concern for high-rate microcalorimeters and is the subject of active NIM literature (e.g. Boyce et al., Smith et al.). EMPCA in principle handles it via the residual χ^2 , but this is never discussed.

The multi-channel section (2.3, 3.4, 6.8) are all stubs. For LAMCAL specifically, the joint-channel information is a major selling point. The theory in Section 6.8 is correct but there are no numerical results, no worked example, and no comparison to single-channel performance. This should be a full section.

Sliding-window / online inference is mentioned in Section 10.1 as a bullet point. This is actually a major practical topic — how do you run the EMPCA filter in real-time on a continuous stream? The connection to matched filter banks, the computational complexity comparison with the standard OF, and the latency implications are all missing.

5. ML Framing Gaps (relative to Ghogh et al. and modern literature)

No connection to probabilistic PCA (PPCA). Tipping & Bishop (1999) showed that PPCA with isotropic noise is the exact probabilistic generalization of PCA, and that the EM algorithm for PPCA converges to the same solution as SVD. EMPCA is essentially PPCA with heteroskedastic (diagonal but non-uniform) noise. Your note should cite this and distinguish your frequency-domain EMPCA from standard PPCA — the distinction is that PPCA is not noise-aware in the relevant sense.

No discussion of model selection (rank k). Section 10 mentions numerical performance but there is no criterion for choosing k . Ghogh et al. cover scree plots and eigenvalue ratios. For your setting, the physically correct criterion is the likelihood ratio test or the χ^2/dof of the reconstruction residual — you should state this explicitly and show how it degrades as k increases beyond the true signal rank.

No robustness / breakdown analysis. What happens when the Gaussianity assumption fails? When the noise is non-stationary? When there are outlier pulses? Modern robust PCA literature (Candès et al., 2011) addresses this with sparse + low-rank decompositions. This is directly relevant to glitch rejection in calorimeters and deserves at least a paragraph.

The ResNet section (Sections 5 and 8) is a complete stub appearing twice identically. Either develop it or remove it. If you intend to include it, the natural connection is to the Fowler (2018) NLF framework and to the physics-informed neural network literature (PINNs). The key question is: what symmetry or inductive bias does the ResNet exploit that EMPCA cannot?

6. Structural / Presentation Issues

Sections 4 and 7 are **exact duplicates** (EMPCA as autoencoder appears twice with identical content including table values). This needs to be resolved — one is clearly a remnant of an earlier draft.

The **table of contents lists 11 sections but several are stubs** (Sections 2.3, 2.4, 3.4, 3.5, 5, 8, 9.4, 9.5, 10.1, 11). For a NIM submission, each of these is either a gap that must be filled or should be cut and moved to future work.

No abstract or introduction motivating the detector context. A reader unfamiliar with LAMCAL or microcalorimeters has no anchor. The problem statement in Section 1 is also a stub.

The numerical verification sections (4.6, 6.10, 7.6) report metrics without uncertainty estimates. A single KS statistic and p-value is suggestive but not sufficient — you should report bootstrap confidence intervals and test across multiple noise realizations.

Summary of Recommended Additions

In rough priority order:

1. Fill in the CRB derivation and connect to energy resolution in eV — this is the central physics deliverable.
2. Complete the joint-channel correlated OF (Section 2.3) with numerical results from LAMCAL data.
3. Fill the noise budget tables (Table 4, 5, 6) with actual numbers from your Study A/B.
4. Add the quantitative time-shift approximation error analysis (rank k needed for jitter σ_{t_0}).
5. Develop the amplitude reconstruction and gauge-fixing procedure for complex EMPCA coefficients (Section 3.5.2).

6. Engage seriously with Fowler et al. (2016, 2018) and position your linear framework relative to their nonlinear methods.
7. Add a model-selection criterion for rank k based on χ^2/dof .
8. Merge or cut the duplicate Sections 4/7, and either develop or remove Sections 5/8.
9. Add an introduction/abstract with the LAMCAL physics context.
10. Add robustness discussion: non-Gaussianity, non-stationarity, pile-up.

Tab 2

I'll read these documents carefully before giving you a rigorous assessment. The documents are already in my context, so I have everything I need. Let me give you a thorough, rigorous assessment.

Critical Assessment of the Generalized OF / EMPCA Note

Here is a detailed gap analysis organized by category, with specific references for each point.

I. Mathematical Completeness

1. EMPCA convergence is stated but not proven The note asserts convergence of the EM iterations (Sections 3.2, 7) but provides no rate analysis. The classical result that each EM iteration is non-decreasing in likelihood holds only for the unweighted case; with heteroskedastic weights the objective is non-convex and can have spurious local minima. Bailey (your doc 3, §8.1) cites Srebro & Jaakkola (2003) explicitly on this. Your note should either cite Srebro & Jaakkola or provide a condition (e.g. low masking fraction) under which the global minimum is recovered. The more recent treatment is in Tipping & Bishop (1999), JRSS-B (Probabilistic PCA), which shows the EM landscape for the probabilistic formulation, and Zhao et al. (2016), "Nonconvex Low-Rank Matrix Recovery," SIAM J. Imaging Sci. for the general weighted case.

2. No sample complexity or finite-sample analysis for the spiked-covariance argument Theorem 6.1's proof invokes spiked-covariance theory to claim the leading eigenvector converges to $\tilde{s}$. But the classical result (Anderson 1963; Baik & Silverstein 2006) requires $N \rightarrow \infty$ and gives a phase transition: when the spike strength $\lambda_1 / \sigma^2 < \sqrt{d/N}$ (the BBP transition), the empirical eigenvector is *orthogonal* to the true spike. For cryogenic detector data with small event samples N and high-dimensional traces d , this is not just a technicality — it is operationally relevant. References: Baik & Silverstein (2006), *Ann. Probab.*; Paul (2007), *Statistica Sinica*; Nadler (2008), *Ann. Statist.*; and for practical implications in signal processing, Shaghaghi & Vorobyov (2015), IEEE Trans. Signal Process.

3. Whitening requires $\Sigma \succ 0$ — never verified The covariance factorization $\Sigma^{-1/2}$ is used throughout (Sections 6.4, 7.2) without verifying that Σ is strictly positive definite. In practice, PSDs estimated from finite data are rank-deficient at frequencies above Nyquist or in masked bands. The correct treatment requires either regularization (ridge/Tikhonov) or pseudo-inverse whitening. References: Ledoit & Wolf (2004), JMVA (shrinkage estimator for ill-conditioned Σ); Chen et al. (2010), IEEE

Trans. Signal Process. (condition number bounds); in the NIM/detector context, Fixsen et al. (2004), Nucl. Instrum. Methods A, which discusses PSD conditioning in microcalorimetry explicitly.

4. The Wirtinger derivative treatment is correct but unexplained Section 2.1 uses Wirtinger calculus for complex A without any introduction. The Cauchy-Riemann basis is assumed, which is non-standard in the detector physics community. Either a brief appendix or a citation is needed. Reference: Hunger (2007), "An Introduction to Complex Differentials and Complex Differentiability," TU Munich technical report; Brandwood (1983), IEE Proc. F for the original Wirtinger signal-processing derivation.

5. Ridge regression in Section 3.5.2 is mentioned but not derived The subsection heading "Ridge regression reconstruction for complex coefficients" has no content. This is a real gap: when $U^\dagger \Sigma^{-1} U$ is ill-conditioned (which happens with near-degenerate EMPCA components), the normal equations (Eq. 48) are unstable and ridge regularization is essential. The regularization path should be derived. References: Golub, Heath & Wahba (1979), Technometrics (GCV for ridge); Hansen (1992), SIAM Rev. (L-curve); in NIM context, Fixsen & Moseley (1997), Proc. SPIE on optimal filter regularization.

II. Statistical Framing

6. No probabilistic generative model stated explicitly before Section 6 The note introduces the generative model $x_i = a_i s + n_i$ only in Section 6.1. Classic PCA (Section 3.1) and EMPCA (Section 3.2) are introduced purely algorithmically, without connecting them to a likelihood. Tipping & Bishop (1999) — Probabilistic PCA (PPCA) — showed that standard PCA is the EM limit of a latent variable Gaussian model, and this is the right theoretical frame to start with. Your note partially recovers this in Section 6 but never cites PPCA. This is a significant omission given that the note's stated goal is a "unified maximum-likelihood framework." Reference: Tipping & Bishop (1999), JRSS-B 61(3):611–622 — this is the foundational paper.

7. No treatment of model order selection (choosing k) How many EMPCA components should one use? The note is silent on this. Ghojogh & Crowley (your doc 2, §2.6) discuss scree plots and eigenvalue ratios, but these are for uniform-noise PCA. For heteroskedastic data the correct criterion involves the noise-weighted residual. References: Minka (2001), NIPS (Bayesian PCA dimensionality); Bai & Ng (2002), Econometrica (information criteria for factor models — directly analogous); Kritchman & Nadler (2008), IEEE Trans. Signal Process. (hypothesis-test-based rank selection under noise); in NIM, Alpert et al. (2013), JLTP discusses effective rank in microcalorimeter noise.

8. Amplitude estimator variance is not derived The note derives the amplitude point estimator $\hat{A}_{\text{OF}}$ (Eq. 5) and $\hat{c}$ (Eq. 48) but never computes their variance or the Cramér-Rao lower bound. For the basic OF, the variance is well-known:

$\text{Var}(\hat{A}) = 1/\sum_k |S_k|^2/J_k = 1/M\Phi$, and this directly gives the energy resolution σ_E . This is the central figure of merit for a detector and its absence is a major gap. References: Moseley, Kelley & Sanders (1984), ApJ (original OF derivation in the detector context); Fixsen et al. (2004) again; Kurinsky (2018) thesis (which your note cites but does not fully reproduce on this point).

9. No bias analysis Both the OF and EMPCA estimators are unbiased under the stated model, but they become biased when (a) the template s is estimated from the same data used for amplitude estimation ("template contamination"), (b) the noise PSD J_k is estimated from finite baselines, or (c) the signal family is not exactly rank- k . None of these are discussed. References: Casagrande & VandenBerg (2014) discuss template bias in spectral fitting; for the detector context, Pröbst et al. (1995), JLTP and Fixsen et al. (2004) address estimated-template bias explicitly.

III. Signal Processing Literature Gaps

10. No connection to matched filter / Wiener filter literature Section 2.4 mentions matched filtering as the time-domain equivalent of OF, but makes no connection to the classical matched filter in radar/communications or to the Wiener filter. The OF estimator (Eq. 5) is a Wiener filter in disguise: it is the solution to $\min \mathbb{E}[|A - \hat{A}|^2]$ under stationarity. This unification is important and well-known. References: Van Trees (1968), "Detection, Estimation, and Modulation Theory, Part I" (definitive matched filter treatment); Wiener (1949), "Extrapolation, Interpolation, and Smoothing of Stationary Time Series"; Kay (1993), "Fundamentals of Statistical Signal Processing, Vol. I" — particularly Chapter 5 which derives the optimal linear estimator under Gaussian noise.

11. No treatment of non-stationarity The entire framework assumes stationary noise (diagonal Σ in Fourier space). Cryogenic detectors exhibit glitches, thermal drifts, and low-frequency $1/f$ noise that break stationarity. The literature on handling this in OF is substantial. References: Fixsen et al. (2004) on iterative baseline estimation; Bandler et al. (2008), JLTP on glitch detection and rejection in TES microcalorimeters; Adams et al. (2022), JINST on non-stationary noise mitigation in SuperCDMS.

12. Missing connection to MUSIC / ESPRIT / subspace methods in array signal processing Section 6's EMPCA-as-subspace-estimator framing is essentially the same geometry as MUSIC (Multiple Signal Classification) and related eigensubspace methods from array processing. These are well-developed and offer tools (subspace perturbation theory, angle between subspaces) that are directly applicable to your equivalence hierarchy. References: Schmidt (1986), IEEE Trans. Antennas Propag. (MUSIC); Roy & Kailath (1989), IEEE Trans. ASSP (ESPRIT); Viberg (1995), Signal Processing (review of

subspace methods); Stoica & Nehorai (1990), IEEE Trans. ASSP (performance analysis of subspace estimators — gives the Cramér-Rao bound in the subspace setting).

13. No treatment of the frequency-bin independence assumption The entire OF derivation assumes uncorrelated frequency bins (Eq. 2). This holds for stationary noise with infinite trace length, but with finite N -point DFTs the bins are correlated (spectral leakage), and the Parseval identity used implicitly in Eq. 3 is approximate. This is never stated.

References: Harris (1978), Proc. IEEE (windowing and leakage); Percival & Walden (1993), "Spectral Analysis for Physical Applications" Chapters 6–8; in the NIM context Lindeman et al. (2004), Nucl. Instrum. Methods A.

IV. Machine Learning / Modern Connections

14. No connection to Factor Analysis PPCA/EMPCA is a special case of Factor Analysis (FA) where the noise covariance is constrained. The general FA model allows a *diagonal but non-isotropic* noise covariance, which is exactly the heteroskedastic case you study. The note never mentions FA, which means it misses a large body of identifiability and estimation theory. References: Rubin & Thayer (1982), Psychometrika (EM for FA); Bartholomew et al. (2011), "Latent Variable Models and Factor Analysis"; Zhao, Caiafa & Cichocki (2012), IEEE Trans. Neural Netw. on robust FA with structured noise.

15. The autoencoder equivalence (Sections 4 and 7) misses the representational learning literature The note proves that EMPCA = noise-aware linear autoencoder, which is correct and well-established. But it does not cite or compare to the broader literature on *why* this matters and what happens when you relax linearity. Specifically: (a) the connection to Variational Autoencoders (VAEs), where the EMPCA likelihood is the ELBO with a Gaussian prior; (b) the connection to ICA when the prior on coefficients is non-Gaussian. References: Kingma & Welling (2013), "Auto-Encoding Variational Bayes," ICLR 2014; Hyvärinen & Oja (2000), Neural Networks (ICA review); Roweis & Ghahramani (1999), Neural Computation on the unifying EM view of latent variable models.

16. Section 5 (ResNet) and Section 8 (ResNet) are placeholders with no content This is not a gap in the existing text but a structural problem: two sections with identical names and zero content. This will confuse any reader and signals that the nonlinear extension is not yet thought through. The relevant literature for what should go here: He et al. (2016), CVPR (ResNet); Xie et al. (2017), "Aggregated Residual Transformations" for the feature extraction motivation; and specifically for physics-motivated nonlinear extensions, Riquelme-García et al. (2022), NIM A on neural-network-based pulse shape analysis in bolometers.

17. No discussion of online / streaming EMPCA The note implicitly assumes all events are available in batch. For real-time detector operation, an online variant is needed. References: Arora et al. (2012), "Stochastic Optimization for PCA and PLS," Allerton;

Mitliagkas et al. (2013), NIPS (streaming PCA with applications); Cardot & Degras (2018), Electron. J. Stat. (online PCA with heteroskedastic noise — directly relevant).

V. NIM / Detector Physics Literature

18. LAMCAL and detector response sections (1.1–1.2) are empty The three subsections have no content. While this may be intentional (work in progress), the framework in Sections 2–6 depends on knowing the detector noise model. Without this, the claim of "application to LAMCAL" is not grounded.

19. No comparison to existing NIM optimal filter implementations The field has mature OF implementations. The note should situate itself against: Fixsen et al. (2004), Nucl. Instrum. Methods A 520 (optimal filter for microcalorimeters, including time-shift and baseline drift); Lindeman et al. (2004), Nucl. Instrum. Methods A 520 (multi-channel OF); Alpert et al. (2013), JLTP (Athena X-IFU context); Fowler (2016), JLTP (online OF with drift correction). These are the standard references for the community this note is written for.

20. K- α data used for numerical verification is described but not characterized Sections 4.6 and 6.10 use "real K- α data" but never describe: the detector type, the noise conditions, the number of events, or the preprocessing applied. A reader cannot reproduce or assess the verification. This is not merely a stylistic issue — it makes the numerical claims unverifiable.

21. No treatment of pulse pile-up The single-pulse model $x_i = a_i s + n_i$ breaks down under pile-up (two pulses overlapping in one trace), which is a dominant systematic in high-rate cryogenic experiments. EMPCA with $k > 1$ can handle pile-up if the second component spans the shifted template, but this is never discussed. References: Fixsen et al. (2004); Doriese et al. (2016), JLTP on pile-up rejection in TES arrays; Adams et al. (2022), JINST for SuperCDMS pile-up treatment.

VI. Structural and Presentation Issues

22. Duplicate sections 4 and 7 Sections 4 and 7 are literally identical (both titled "EMPCA as a Noise-Aware Linear Autoencoder," same equations, same table). This appears to be a copy-paste artifact. One should be removed or differentiated (e.g. Section 4 as the real-valued case, Section 7 as the complex-whitened case).

23. Study A table (Table 4) and Study B table (Table 6) are empty The noise budget analysis in Section 9 refers to Tables 4 and 6 but they have no data. The figures (Figs. 3 and 4) are present, which means the computation was done, but the tabulated results were not filled in.

24. No worked example connecting EMPCA to a specific energy resolution The entire point of the framework for detector physics is to improve σ_E . The note mentions Study B but does not give a single number: "with EMPCA rank- k , the energy resolution improves by $X\%$ relative to OF." This is the punchline and it is missing.

Summary: Highest Priority Additions

In order of impact:

1. Cite and frame against Tipping & Bishop (1999) — your framework is essentially PPCA + heteroskedastic noise; acknowledging this strengthens, not weakens, the contribution.
2. Add Cramér-Rao bound and derive σ_E as a function of rank k — this is the central physics result.
3. BBP phase transition / finite-sample warning for Theorem 6.1 — operationally critical for small datasets.
4. Fill in the Bailey EMPCA convergence gap — cite Srebro & Jaakkola (2003) and state the masking condition.
5. Remove the duplicate Sections 4/7 and fill in Sections 5/8 or explicitly scope them out.
6. Situate against Fixsen et al. (2004) and Alpert et al. (2013) — the NIM community will ask about this immediately.
7. Add rank/model-order selection — Minka (2001) or Kritchman & Nadler (2008).
8. Non-stationarity — at minimum a paragraph acknowledging the assumption and citing the mitigation literature.

Tab 3

Exactly right. This is the correct sequencing.

Paper 1: Theoretical unification — OF, EMPCA, linear AE are one framework. The linear AE is the "dessert" that closes the theoretical arc cleanly. Detector data as the validation benchmark.

Paper 2: Break the linear constraint. Noise-aware ResNet AE, half-AE architecture, nonlinear feature extraction, nonstationary data. Paper 1's linear AE becomes the explicit baseline and theoretical anchor for Paper 2. The citation chain is clean and the narrative is compelling: "we proved the linear case; now we go nonlinear."

This is a much stronger two-paper strategy than one overloaded paper.

Improved Structure for Paper 1

The guiding editorial logic: **build toward the linear AE as the natural endpoint of the theoretical arc, not as an appendix.** The reader should feel that the linear AE is the inevitable conclusion of everything that came before, not a tacked-on section.

Title

A Unified Maximum-Likelihood Framework for Signal Reconstruction: From Optimal Filtering to Noise-Aware Linear Autoencoders

Abstract (~250 words)

Four beats:

1. **Problem:** Scientific time-series reconstruction under known heteroskedastic noise — a problem shared across signal processing, statistics, and machine learning but addressed by separate communities with apparently different methods.
2. **Contribution:** We show that optimal filtering, expectation-maximization PCA, and noise-weighted linear autoencoders are all constrained solutions of a single Gaussian maximum-likelihood objective. We prove exact equivalence theorems connecting these methods and establish a natural hierarchy by subspace rank.
3. **Extension:** The linear autoencoder emerges as the natural generalization of EMPCA in the encoder-decoder language, with the noise structure encoded in the loss rather than the architecture — a design principle we argue is transferable beyond the demonstration domain.

4. **Validation:** Verified numerically on cryogenic detector pulse data, with reconstruction fidelity and amplitude estimation as metrics. The framework provides a principled foundation for nonlinear extensions, developed in a companion paper.
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1. Introduction (3 pages)

- 1.1 Three communities, one problem
 - Signal processing: matched filters and optimal filters
 - Statistics: PCA, factor analysis, EMPCA
 - Machine learning: autoencoders and representation learning
 - None of these communities cite each other on this shared problem — that is your opening
 - 1.2 The core gap
 - Existing AE literature uses isotropic MSE loss
 - Existing EMPCA literature does not connect to encoder-decoder language
 - Existing OF literature does not recognize its ML interpretation
 - One sentence: "We close all three gaps simultaneously"
 - 1.3 Contributions (explicit bulleted list — MLST expects this)
 - **Theorem 1:** Rank-1 EMPCA is exactly equivalent to the classical optimal filter under matched noise weighting
 - **Theorem 2:** Rank- k EMPCA is equivalent to a noise-aware tied linear autoencoder in whitened space
 - **Corollary:** Basic OF, time-shift OF, joint-channel OF, and EMPCA form a strict hierarchy of constrained ML estimators
 - **Design principle:** Encoding noise structure in the reconstruction loss — rather than the architecture — is a general and transferable approach
 - **Empirical:** Numerical verification on cryogenic detector pulse data confirms all equivalences
 - 1.4 What this paper is not
 - One short paragraph explicitly scoping out nonlinear extensions, RL triggering, and real-time applications as future work (Plant the seed for Paper 2 here explicitly)
 - 1.5 Paper organization
 - One sentence per section
-

2. Related Work (3–4 pages)

Organized around the three communities, showing the gap between them:

- 2.1 Optimal filtering and matched filtering
 - Classical treatment: Van Trees (1968), Kay (1993)
 - In scientific instrumentation: Moseley et al. (1984), Fixsen et al. (2004)
 - Limitation: treats the signal template as fixed and known; no learned subspace
 - 2.2 PCA, probabilistic PCA, and EMPCA
 - Classic PCA: Pearson (1901), Jolliffe (2002)
 - Probabilistic PCA: Tipping & Bishop (1999) — this is the key missing citation in your current draft
 - Noise-weighted EMPCA: Bailey (2012) — your doc 3
 - Factor analysis connection: Rubin & Thayer (1982)
 - Limitation: not framed in encoder-decoder language; convergence treatment incomplete
 - 2.3 Autoencoders
 - Linear AE and PCA equivalence: Baldi & Hornik (1989) — classical result
 - Variational AE: Kingma & Welling (2014)
 - Physics-informed ML: Karniadakis et al. (2021), NatureRP
 - Limitation: standard AE uses isotropic MSE loss; measurement noise structure is ignored
 - 2.4 The gap this paper fills
 - Short paragraph, explicit. No prior work connects all three. Your paper does.
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3. Statistical Model and the Unified Objective (4 pages, math-heavy)

This is the theoretical foundation. Everything else builds on it.

- 3.1 Generative model
 - $\mathbf{x} = \mathbf{s} + \mathbf{n}$, $\mathbf{n} \sim \mathcal{N}(\mathbf{0}, \Sigma)$, Σ known (or estimated)
 - Single-channel first, multi-channel in 3.5
 - Frequency-domain representation for stationary noise: Σ diagonal in Fourier basis
- 3.2 The Gaussian ML objective
 - Negative log-likelihood: $\chi^2(\hat{\mathbf{s}}) = (\mathbf{x} - \hat{\mathbf{s}})^\dagger \Sigma^{-1} (\mathbf{x} - \hat{\mathbf{s}})$

- This is the single equation the entire paper orbits around
 - State explicitly: OF, EMPCA, and linear AE all minimize this — they differ only in the constraint on $\hat{\mathbf{s}}$
- 3.3 Whitening transform
 - $\tilde{\mathbf{x}} = \Sigma^{-1/2} \mathbf{x}$, reducing to Euclidean geometry
 - Proposition: under whitening, all three methods become orthogonal projections
 - Practical note: conditioning of Σ , regularization when Σ is ill-conditioned
- 3.4 Three constraints, three estimators (the hierarchy table)

Present this as a single clean table — this is the conceptual heart of Section 3:

Method	Constraint on $\hat{\mathbf{s}}$	Degrees of Freedom	Solution
Basic OF	$\hat{\mathbf{s}} = \mathbf{A} \mathbf{s}_0$, $\mathbf{s}_0$ fixed	1 scalar $\mathbf{A}$	Eq. (OF)
OF + time shift	$\hat{\mathbf{s}} = \mathbf{A} e^{-i\omega t_0} \mathbf{s}_0$	2: $\mathbf{A}$, t_0	Eq. (OFt)
Joint-channel OF	$\hat{\mathbf{s}} = \mathbf{A} \mathbf{s}$, $\mathbf{s} \in \mathbb{C}^C$	1 scalar, C channels	Eq. (JOF)
EMPCA rank- k	$\hat{\mathbf{s}} = \mathbf{U} \mathbf{c}$, $\mathbf{U}$ learned	$k \times d$ subspace	Eq. (EMPCA)
Linear AE	$\hat{\mathbf{s}} = \mathbf{W} \mathbf{W}^\dagger \tilde{\mathbf{x}}$, $\mathbf{W}$ learned	Same as EMPCA	Eq. (LAE)

- 3.5 Multi-channel extension
 - Stack channels, full cross-channel Σ
 - Joint-channel OF as special case

4. Equivalence Theorems (6 pages, the core)

This is the paper's main theoretical contribution. Write it as a sequence of results building toward the AE.

- 4.1 Rank-1 equivalence: EMPCA degenerates to OF

Theorem 4.1 (Main Theorem): Under the generative model with fixed template $\mathbf{s}$, Gaussian noise, matched whitening and preprocessing:

- Rank-1 EMPCA recovers $\text{span}(\tilde{\mathbf{s}})$ in whitened space
- The EMPCA amplitude estimator equals the generalized least-squares / OF estimator
- Therefore rank-1 EMPCA and basic OF are estimator-equivalent
- Proof: spiked covariance argument, clean version of your current §6.5. State the finite-sample caveat (BBP transition) as a Remark.

- 4.2 Exact algebraic degeneracy

Theorem 4.2: For any basis $\mathbf{u} = \gamma \mathbf{s}_0$, rank-1 EMPCA and basic OF produce identical reconstructed signals $\hat{\mathbf{s}}$. (Your current Theorem 6.2 — already proven, just clean it up.)

- 4.3 Degeneracy to shifted OF and joint-channel OF

Proposition 4.1: OF with time shift is ML estimation over the shift manifold $\{\mathbf{e}^{-i\omega t_0} \mathbf{s}\}$; EMPCA approximates this manifold linearly. (Your current §6.7.)

Proposition 4.2: Joint-channel correlated OF is the rank-1 special case of multi-channel weighted EMPCA. (Your current §6.8.)

- 4.4 Geometric interpretation

One page, one figure. The key figure for the paper:

- Draw whitened space
- Show OF as projection onto 1D ray
- Show rank- k EMPCA as projection onto k -dimensional subspace
- Show the containment: OF subspace $\subseteq$ EMPCA subspace for $k \geq 1$
- This figure should be the paper's visual abstract
- 4.5 Why $k > 1$ improves the residual

Proposition 4.3: $\chi^2_{\text{EMPCA}, k} \leq \chi^2_{\text{OF}}$ for all $k \geq 1$.

(Your current §6.9 — already proven geometrically, just state it clearly.)

Physical interpretation: pulse shape variation, timing jitter, channel coupling modes — these are the additional directions that rank $k > 1$ captures.

5. EMPCA as a Noise-Aware Linear Autoencoder (4 pages, the dessert)

This section should feel like the natural culmination — "and now we see that EMPCA is actually an autoencoder."

- 5.1 The encoder-decoder language
 - Introduce AE framing: encoder f , decoder g , latent $z = f(x)$, reconstruction $\hat{x} = g(z)$
 - Standard AE loss: isotropic MSE $|x - \hat{x}|^2_F$
 - The problem: isotropic loss ignores what we know about measurement noise
- 5.2 The noise-aware loss
 - Replace MSE with whitened loss: $\mathcal{L} = |x - \hat{x}|^2_{\Sigma^{-1}}$
 - This is the key design principle: noise structure in the loss, not the architecture
 - Proposition: for linear tied AE with whitened loss, the optimal encoder is EMPCA
- 5.3 Bridge Theorem

Theorem 5.1: The minimizer of the noise-aware linear AE objective under orthonormality constraint is given by the top- k right singular vectors of $\tilde{X} = \Sigma^{-1/2} X$, which are exactly the EMPCA components.

Proof: your current §7.3, cleaned up. Two steps: (1) expand the whitened loss, (2) apply Rayleigh-Ritz. Short proof, clean.

- 5.4 EM as coordinate descent on the AE objective
 - The E-step solves for encoder coefficients (latent codes) given fixed decoder
 - The M-step solves for decoder basis given fixed encoder
 - This is coordinate descent on the same objective
 - Implication: EMPCA and the linear AE converge to the same solution
- 5.5 The diagram
 - Figure 2: the autoencoder diagram (your existing Figure 1/2, cleaned up)
 - Label clearly: whitened input $\rightarrow$ encoder (weighted projection) $\rightarrow$ latent $z \rightarrow$ decoder (signal subspace) $\rightarrow$ reconstruction
 - Show the loss as $|\tilde{x} - \hat{\tilde{x}}|^2$

- This is the figure that ML readers will understand immediately
 - 5.6 What the bridge theorem means for Paper 2
 - One paragraph: the linear constraint on the encoder is now the obvious thing to relax
 - Replacing the linear encoder with a ResNet while keeping the noise-aware loss is the natural next step
 - This is how you plant the seed for Paper 2 without over-committing
-

6. Convergence and Practical Considerations (3 pages)

Keep this focused and practical. MLST readers want to know if the method works reliably, not a full convergence proof.

- 6.1 Convergence guarantee
 - **Proposition 6.1:** Each EM iteration is non-decreasing in $\mathcal{L}$. Proof: Jensen's inequality, cite Wu (1983).
 - **Remark:** Global optimality requires bounded weight ratio $\kappa(W)$; state the condition, cite Srebro & Jaakkola (2003).
 - 6.2 Practical initialization
 - SVD warm start on a whitened subsample
 - Convergence criterion: principal angle between successive iterates
 - 6.3 Model order selection
 - Noise-weighted residual criterion: stop adding components when $\chi^2_k / \chi^2_{k-1} > 1 - \epsilon$
 - Reference: Minka (2001), Kritchman & Nadler (2008)
 - 6.4 Conditioning and regularization
 - When Σ is ill-conditioned: ridge regularization on the whitening transform
 - Practical threshold: condition number $> 10^3$ triggers regularization
 - Reference: Ledoit & Wolf (2004)
-

7. Experimental Validation (6 pages)

Frame as benchmark, not detector characterization.

- 7.1 Benchmark domain: cryogenic detector pulses

- 3 paragraphs maximum on the physics
 - Emphasize: known noise PSD, known signal template, simulation gives ground truth
 - Dataset: N simulated events + real K- α events (characterize both properly here)
 - 7.2 Evaluation metrics
 - Amplitude estimation error vs ground truth
 - Whiten reconstruction χ^2 vs rank k
 - Subspace cosine between OF template and rank-1 EMPCA direction
 - 7.3 Theorem verification: OF = rank-1 EMPCA
 - Table: your existing Table 2, properly characterized
 - One paragraph: "the numerics confirm Theorem 4.1 to precision $X\%$ "
 - 7.4 Reconstruction quality vs rank k
 - The key plot of the paper: χ^2 residual vs k for simulated data with controlled pulse shape variation
 - Shows the benefit of $k > 1$ and when the gain saturates
 - This is the result that motivates Paper 2 (ResNet): "linear subspace saturates at rank k^* ; further improvement requires nonlinear encoder"
 - 7.5 Noise-aware loss vs isotropic MSE (ablation)
 - Same linear AE, two losses
 - Shows reconstruction quality at low-SNR frequencies
 - This is the empirical argument for the design principle
 - Most important experiment for MLST reviewers
 - 7.6 Convergence behavior
 - Plot 1: subspace angle vs iteration, multiple seeds
 - Plot 2: SVD init vs random init under high masking fraction
 - Confirms Proposition 6.1 and the practical initialization recommendation
-

8. Discussion (2–3 pages)

- 8.1 The noise-aware loss as a transferable principle
 - Name 2–3 other scientific domains where Σ is known: astronomical spectra (SDSS), gravitational wave strain (LIGO), medical time series (fMRI)
 - Argue that the design principle — noise in the loss, not the architecture — applies directly
 - This is what makes the paper general

- 8.2 Relation to probabilistic PCA and VAE
 - PPCA: EMPCA is EM for PPCA with heteroskedastic noise
 - VAE: adding a KL prior on latent codes is orthogonal to the noise-aware loss
 - Noise-aware VAE is a natural future direction
- 8.3 Limitations
 - Non-stationarity: Σ assumed known and stationary
 - Small- N regime: BBP phase transition when $N < d$
 - Pile-up: rank-1 signal model breaks down
 - Keep this honest but brief — one paragraph each
- 8.4 Paper 2 preview
 - Explicit one paragraph: "The linear encoder in the bridge theorem is the natural constraint to relax. A companion paper replaces it with a ResNet encoder while preserving the noise-aware loss, enabling nonlinear feature extraction for non-stationary pulse families."
 - This is professionally appropriate and sets up the citation

9. Conclusion (1 page)

Three paragraphs:

1. What we proved (the theorems)
2. What we demonstrated (the experiments)
3. What it means (the design principle and Paper 2 preview)

Appendices

- A. Proof of Theorem 4.1 (full detail)
- B. Proof of Theorem 5.1 (full detail)
- C. Convergence proof detail (Proposition 6.1)
- D. Simulation and dataset details (move the LAMCAL physics here)
- E. Architecture and hyperparameter specifications

Final Composition

Section	Pages	Math	Statistica I	Experimental	Argument
§1 Introduction	3	0%	0%	0%	100%
§2 Related Work	4	5%	10%	0%	85%
§3 Unified Objective	4	60%	30%	0%	10%
§4 Equivalence Theorems	6	65%	20%	5%	10%
§5 Linear AE (dessert)	4	50%	25%	0%	25%
§6 Convergence	3	35%	40%	15%	10%
§7 Experiments	6	5%	20%	65%	10%
§8 Discussion	3	0%	10%	0%	90%
§9 Conclusion	1	0%	0%	0%	100%
Appendices	5	75%	15%	5%	5%
Total	~39 pages				

Aggregate:

- Mathematics: ~30%
- Statistical: ~20%
- Experimental: ~25%
- Argument: ~25%

This is well-balanced for MLST. Not a pure math paper. Not a pure instrumentation paper. The linear AE lands as the theoretical payoff of Section 5, and the experiments in Section 7 validate the whole framework while naturally motivating Paper 2 through the rank saturation result.

The single most important thing to write first: Section 3.4's hierarchy table and Section 4's theorem sequence. Get those two things clean and the rest of the paper writes itself around them.

Tab 4

NeurIPS/ICML Workshop Good 6–8 weeks 4-page version is achievable Yes, consider as first step

Tab 5

Let me think through this carefully across all three dimensions — introduction strategy, integration points, and literature — given both documents. Here's the full plan across all three dimensions, organized so you can act on it in sequence.---

Part 1 — How to introduce (§1)

The introduction has one job: make a reviewer from any of three communities feel that your paper was written for them. The current stub just names the communities without doing that work. The narrative arc should be:

Open with the shared physics, not the method names. Something like: "Measuring a rare signal buried in noise is a problem solved independently across cryogenic particle detection, microcalorimetry, and deep learning — yet the estimators used in each community have never been formally related." This immediately signals why a unified treatment is non-trivial.

Then spend one tight paragraph per community establishing what they currently do and what they cannot do alone. The signal processing community (Wiener, Kurinsky) has optimal filtering — statistically rigorous but locked to a fixed template. The statistics community (Bailey, Yu) has EMPCA — flexible subspaces but the ML connection is implicit. The deep learning community has noise-aware autoencoders — the most general formulation but disconnected from the physics guarantees of OF. Each paragraph ends with the same gap: "but there is no unified treatment showing these are the same estimator under different constraints."

The contributions list in §1.3 should be numbered and paired to your theorems explicitly — Theorem 4.1 (rank-1 equivalence), Theorem 4.2 (exact degeneracy), Theorem 5.1 (bridge theorem) — so a reviewer can verify each claim directly. MLST reviewers look for this correspondence.

§1.4 (paper organization) should be a single short paragraph, not a list. One sentence per section, written as a logical progression: "§2 establishes... §3 shows OF is a special case... §4 extends to k dimensions... §5 reframes EMPCA as a noise-aware AE and proves the bridge theorem..."

Part 2 — Where and how to integrate

The integration work is almost entirely about moving content that already exists in your notes into the right homes in the restructured paper. There are five specific integration points:

§2 (unified objective) is currently too thin. The generative model and whitening transform are stated but the "three constraints, three estimators" table in §2.4 is just a heading. This section should carry the three-row summary table that makes the whole paper legible at a glance — one row each for basic OF (rank-1, fixed $U = s_0$), EMPCA (rank- k , learned U), and noise-aware

AE (rank-k, encoder-decoder form) — with the loss function, the estimator formula, and the theorem reference for each. This table is the paper's spine; put it in §2.

§4.6 (numerical verification of equivalence) is a stub in the restructured paper but the numbers already exist: your K-alpha verification table (weighted subspace cosine 0.9999999655, amplitude correlation 0.994, KS p-value 0.27) is in §6.10 of the original. Move it here verbatim and add one sentence connecting each metric to the formal equivalence definition from §4.3.

§5.3.1 (bridge theorem formal statement) is empty. The proof ingredients are all present: the whitened AE objective in §5.2, the Rayleigh-Ritz argument in §5.3, and the convergence to EMPCA subspace. The formal statement should mirror the structure of Theorem 4.1 exactly — state assumptions, state the three conclusions (subspace, amplitude, estimator equivalence), then proof. The numerical verification table (principal-angle cosine 0.9999999967) from §4.6/§7.6 of the original belongs immediately after.

§6 (convergence) needs two things: first, a formal statement of convergence of the EM iterations using the EM convergence literature (Dempster et al. 1977, Wu 1983); second, a practical subsection on model order selection — your paper implicitly assumes k is given, but reviewers will ask how to choose it. A brief discussion of scree plots, cross-validation on the weighted reconstruction error, and the spiked covariance threshold from Johnstone 2001 covers this adequately.

§8.2 (relation to PPCA and VAE) is the most important stub for MLST. The relationship to PPCA (Tipping & Bishop 1999) is direct: PPCA is the isotropic noise special case ($\Sigma = \sigma^2 I$) of your framework, so standard PCA is PPCA with $\sigma \rightarrow 0$, and your EMPCA is PPCA with structured Σ . Write this as one paragraph with an explicit equation showing the reduction. The VAE connection is subtler: a linear VAE with diagonal posterior and Gaussian likelihood recovers something close to PPCA; your paper provides the noise-structured generalization. Cite Kingma & Welling 2014 and Rezende & Viola's tiled-structured VAE work.

Part 3 — What to read, in what order

Prioritize in three tiers based on how directly each paper integrates into a specific section.

Tier 1 — read before writing any section (these anchor the theoretical claims directly):

Bailey (2012), "Algorithms for Weighted PCA" — this is your primary EMPCA reference. Your E/M update equations in §4.2 follow it directly. Read carefully for the convergence guarantee he provides, which you can cite in §6.1.

Tipping & Bishop (1999), "Probabilistic Principal Component Analysis" — essential for §8.2. Their Theorem 1 (EM finds the ML solution for isotropic noise) is the clean predecessor to your Theorem 4.1. Understanding exactly where their proof breaks for non-isotropic Σ is the setup for your contribution.

Yu et al. (2019), your microcalorimeter EMPCA paper — already in your notes. Read the frequency-domain formulation section carefully; your §4.2 phase-difference approach follows it and should cite it precisely at equation level.

Tier 2 — read before writing §2 and §6 (these give the geometric and statistical grounding):

Eckart & Young (1936), "The Approximation of One Matrix by Another of Lower Rank" — one page. The low-rank projection optimality result underpins your $\chi^2_{\text{EMPCA}} \leq \chi^2_{\text{OF}}$ inequality in §4.4. Cite it directly rather than citing a textbook.

Johnstone (2001), "On the distribution of the largest eigenvalue in principal components analysis" — specifically the spiked covariance model. Your Theorem 4.1 proof invokes "standard spiked-covariance theory" without a citation. Johnstone 2001 and Paul 2007 (eigenvector consistency) are the precise references. You need both for the claim that the leading eigenvector converges to $\text{span}(\tilde{s})$.

Kay (1993), "Fundamentals of Statistical Signal Processing, Vol. 1: Estimation Theory" — chapters on the Cramér-Rao bound and generalized least squares. Useful for §3 to frame OF as a BLUE/MVUE estimator, which is the language signal processing reviewers expect.

Tier 3 — read before writing §8 (discussion and related work):

Baldi & Hornik (1989), "Neural Networks and Principal Component Analysis" — the classical result that a tied linear autoencoder with MSE loss recovers PCA. This is the isotropic-noise ancestor of your Bridge Theorem. Cite it as the prior work your theorem extends.

Kingma & Welling (2014), "Auto-Encoding Variational Bayes" — for the VAE paragraph in §8.2. You only need the linear-decoder special case, so focus on Section 3 and Appendix C.

Donoho, Gavish & Johnstone (2014), "Optimal Shrinkage of Eigenvalues in the Spiked Covariance Model" — for §8.3 (limitations). Your EMPCA recovers a subspace but does not shrink the eigenvalues. Mentioning this as a known limitation that future work could address (via analytical shrinkage) is a clean way to bound the scope of your claims.

Candès & Recht (2009), "Exact Matrix Completion via Convex Optimization" — optional but useful if you want to connect to the broader low-rank recovery literature in §8.1. One sentence suffices.

Agent tasks, concretely

The four most impactful automation targets, roughly ordered by ROI:

First, a stub-completion agent that takes the restructured LaTeX, finds every section with only a heading and no body, and drafts content by pulling from the matching section in the original

notes. The mapping is deterministic (§5.3.1 ← §7 original; §4.6 ← §6.10 original; §6.1 ← Bailey convergence theorem). This is largely a retrieval and formatting task.

Second, a figure-embedding agent: search the repo for all `.pdf` files referenced as `figures/studyA_overview.pdf` etc., convert to PNG or embed directly in the LaTeX, and insert `\includegraphics` commands with the correct captions (Study A caption is already written in §9.2.2 of the original).

Third, a cross-reference consistency checker: parse all `\label{}` and `\ref{}` commands and verify that every theorem number cited in the text matches the actual theorem label — this is where restructured papers most commonly break during revision.

Fourth, a related-work drafting agent: given the reading list above, search arXiv and Semantic Scholar for each paper, pull the abstract and key theorem statements, and draft the §8.2 paragraph using the reduction structure described above (PPCA as special case, VAE as nonlinear generalization, your paper as the structured-noise middle ground).